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Iodine Spectrometer

Abstract

The aim of this experiment was to analyse the band structure of the diatomic molecule iodine by determining molecular constants such as the fundamental vibrational frequency, the anharmonicity constant and the dissociation energy, which were then compared with accepted literature results falling within reasonable agreement. The light of a tungsten lamp transmitted through a cell containing iodine vapour was analysed obtaining its spectrum and ultimately the Birge-Sponer plot from which the molecular constants ($\omega_e = 128.3 \pm 1.9\,\mathrm{cm}^{-1}$, $\omega_e' X_e' = 0.908 \pm 0.039\,\mathrm{cm}^{-1}$, $D_e' = 4497 \pm 213\,\mathrm{cm}^{-1}$) pertaining the electronic state B were extracted from. A good linear fit was obtained for the Birge-Sponer plot showcasing the linear relationship between the first differences of the bands and their vibrational quantum numbers v'. The B band of the molecule was successfully studied and its Morse potential as a function of internuclear radius was obtained allowing thorough visualization of the molecule's vibrational structure.

Contents

1 Introduction

Molecular spectra arise from transitions between rotational, vibrational, and electronic energy levels[1]. When molecules absorb or emit radiation, simultaneous changes in these levels occur, producing characteristic band systems in the ultraviolet and visible regions[1]. Iodine is a particularly useful case: its long absorption series contains bands corresponding to transitions from the electronic ground state X to the excited state B, yielding a rich and accessible band spectrum in the visible range (400–650 nm)[2].

The aim of this experiment was to analyse the vibrational structure of iodine by determining molecular constants such as the fundamental vibrational frequency, the anharmonicity constant, the dissociation energy, and the force constant, which were then compared the with accepted literature results.

This was done by passing the continuous light of a tungsten lamp through a glass cell containing iodine vapour after which, the transmitted spectrum was recorded using a *Baird & Watts Monospek 1000* spectrometer. After calibration, the absorption bands were analysed and a plot of the first differences $\Delta^1 G(v)$ against the vibrational quantum number v was constructed, from which the molecular constants of iodine were extracted.

2 Theory

2.1 Molecular bands and spectral structure of I_2

There exists three types of band systems in molecules: rotational (due to the rotation of the molecules around its centre), vibrational (due to the vibration of the atoms composing the molecule) and electronic (due to the electronic transitions within the molecule)[1]. Absorption or emission of radiation by the molecule results in changes on their rotational, vibrational and/or electronic energies which can be represented in the molecule's spectrum[1]. Due to the large amount of energy involved in electronic transitions, these are often accompanied by a change the molecule's rotation and vibrational bands leading to broader absorption or emission bands[1].

Figure 1 shows a plot of potential energy as a function of interatomic distance r. Curves X, A and B represent the different electronic states with X usually representing the ground state and A, B, C... representing the excited states in order of increasing energy[3]. The A state results in a spectrum in the red side of the visible region which will not be used in this experiment[3]. Instead, states X and B will be the ones studied.

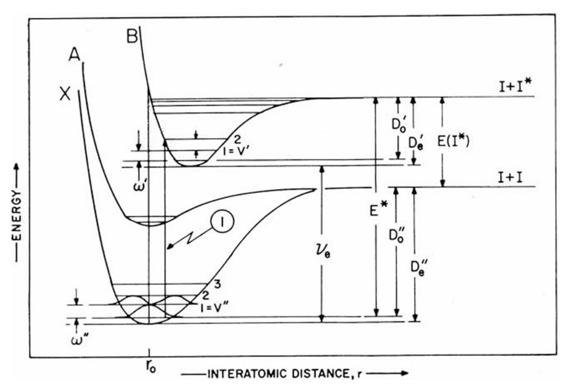

Figure 1: Representation of molecular band structure - potential energy as a function of internuclear radius r [3]

Vibrational levels, represented as $v = 0, 1, 2...$ are associated to the characteristic vibrational energies ω which are grossly exaggerated in the diagram[3]. Notation like v' and v'' or ω' and ω'' might be used to differentiate between the parameters associated with the X and B states respectively.

D_0 indicates the dissociation energy of the state whereas D_e represents the *extrapolated* dissociation energy [3]. This is the energy at which a molecule reaches its dissociation limit and at which the spectral bands disappear[1]. Past this point, the energy differences amongst vibrational levels is zero[1].

A transition between two different vibrational levels belonging to different electronic states are represented as vertical lines (arrow 1 in Fig. 1). This transition can be written as $(B, v' = 2 \leftarrow X, v'' = 0)$[3].

Each transition appears in the spectrum of the molecule as a "bump" forming, as a whole, an image like the one shown in Figure 2. Region (2) shows the wavelengths at which not only the vibrational series $v'' = 0$ are noticeable but also the ones corresponding to $v'' = 1$. Region (3) is where dissociation is reached and the distance between vibrational levels is zero. Meanwhile, region (4) indicates the start of where transitions from the X state to the A state would be, transitions which will not be used in this experiment[3].

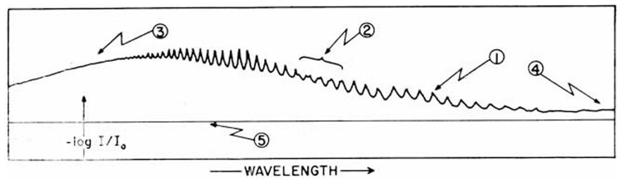

Figure 2: Spectrum of I_2 in air in the 400-650 nm range. (1) A band head. (2) Region where (v', 1") bands appear in the I_2 spectrum. (3) Region in which the convergence limit is located. (4) Start of the (A $\leftarrow$ X) near-infrared bands. (5) "Base" line.[3]

Figure 3 zooms into the central region of the spectrum. Where absorption lines caused by vibrational series $v'' = 1$ and $v'' = 2$ can be appreciated and what are refereed to as sections (2), (3) and (4) 2 can be seen (note here the wavelength is decreasing left to right).

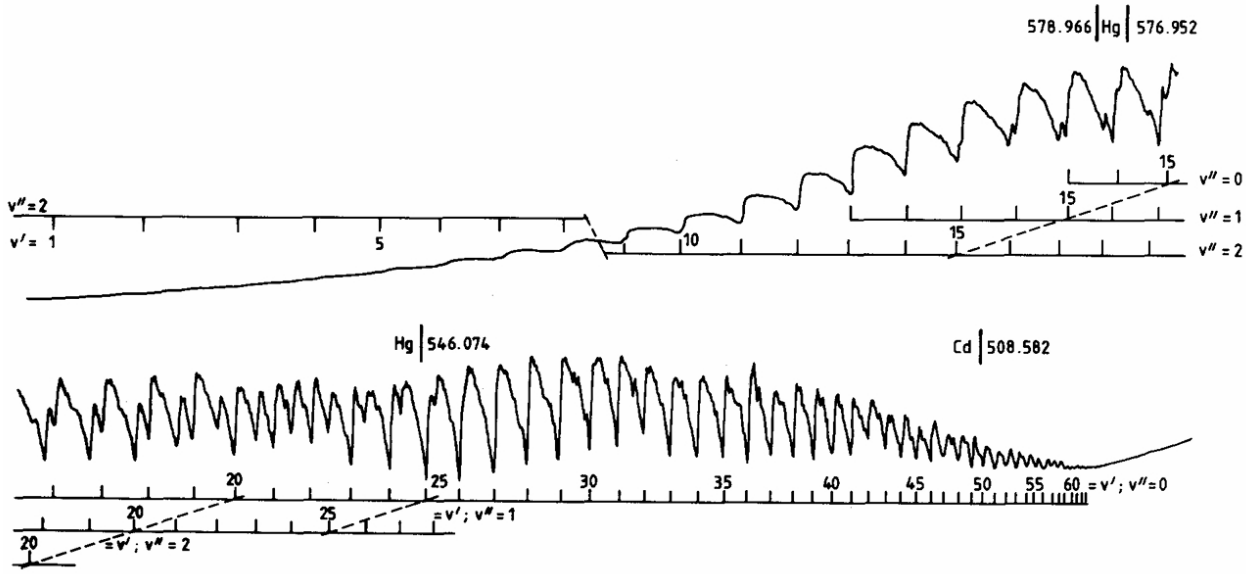

Figure 3: Iodine's visible absorption spectra along with reference lines from mercury and cadmium lamps as well as the three vibrational series $v'' = 0, 1, 2$[2].

2.2 Vibrational structure and numerical determination of the dissociation energy

Rotational levels were not included in Figure 1 as their energy cannot be resolved with this experiment[3]. It is then safe to define the total energy of the molecule as

$$E = T_e + G(v) \tag{1}$$

where T_e and $G(v)$ are the electronic and vibrational energies respectively[1]. $G(v)$ can be written as follows:

$$G(v) = \omega_e(v+1/2) - \omega_e X_e(v+1/2)^2 + \dots \tag{2}$$

with ω_e and $\omega_e X_e$ being the fundamental vibrational frequency and the anharmonicity constant[1].

These parameters can be determined by examinating the spacing between successive vibrational bands:

$$\Delta^1 G(v) = \Delta\nu \tag{3a}$$

$$\Delta^2 G(v) = \Delta^1 G(v) - \Delta^1 G(v+1) \tag{3b}$$

where $\Delta^1 G(v)$ and $\Delta^2 G(v)$ are respectively denominated the first and second differences[1]. $\Delta\nu$ on the other hand, is the difference in the radiation frequencies between transitions from v'' to v' levels when absorption occurs[1]. These frequencies can be defined as follows[1]

$$\nu = (T'_e - T''_e) + G(v) - G(0) \tag{4}$$

leading to

$$\Delta\nu = G(v+1) - G(v)$$

By plugging this and Equation 2 into Equation 3, the first and second differences can be rewritten as

$$\Delta^1 G(v) = \omega_e - 2(v+1)\omega_e X_e \tag{5a}$$

$$\Delta^2 G(v) = 2\omega_e X_e \tag{5b}$$

Once dissociation is reached, the spacing between band will be zero, i.e.

$$\omega'_e - 2(v'+1)\omega'_e X'_e = 0$$

resulting in the determination of the vibrational quantum number $v = v_m$ at which the dissociation limit is reached[1]:

$$v_m = (1/2X_e) - 1 \tag{6}$$

This allows the calculation of the dissociation energy[1]:

$$D_e = G(v_m) \tag{7}$$

2.3 The Birge-Sponer plot

The Birge-Sponer plot consists of plotting the first differences $\Delta^1 G(v)$ against the vibrational quantum number (v). This unveils a linear relationship between the two with a slope equal to twice the negative anharmonicity constant (-$2\omega_e X_e$) and an intercept at $v = -1$ equal to ω_e[1]. However, for large values of v ($v' > 50$[2]) the cubic and quadratic terms of the $G(v)$ expansion make the curve deviate from the linear relationship making the decrease in $\Delta^1 G(v)$ steeper as v increases, just as show in Figure 4[1]. Once ω_e is obtained, the force constant can be calculated as

$$k_e = 4\pi^2 c^2 \mu \omega_e \tag{8}$$

where c is the speed of light and μ is the reduced mass of the molecule[1].

The straight line can be extrapolated to find the vibrational quantum number $v = v_m$ and the dissociation energy D_e as the area under the curve[1].

Once these molecular constants have been obtained, the Morse potential for the given electronic state can be obtained as

$$U(r - r_e) = D_e(1 - e^{-\beta(r-r_e)})^2 \tag{9}$$
$$\beta = \omega_e(2\mu c/D_e h)^{1/2}$$

where r_e is the equilibrium internuclear distance[1]. $r_e = 3.016 \times 10^{-8}$ cm for iodine's B electronic state and $r_e = 2.667$ for its X state[4].

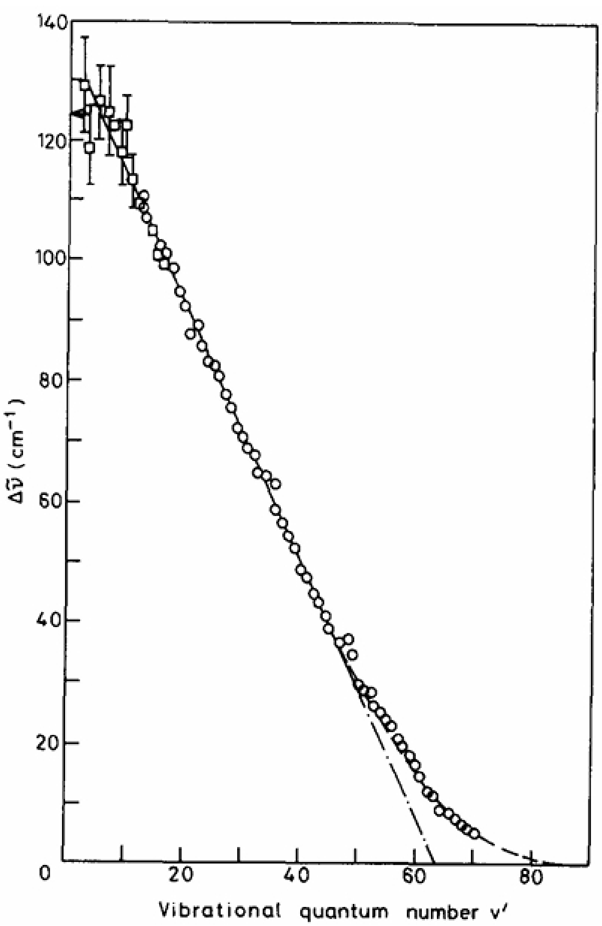

Figure 4: Birge-Sponer plot of the B-X system of molecular iodine with the differences $\Delta\tilde{\nu} = \Delta^1 G(v')$ being plotted in cm^{-1} against the vibrational quantum number v'[2].

3 Methodology

3.1 Apparatus and set up

The apparatus, as seen in Figure 5, consisted on a *Baird & Watts Monospek 1000* powered by a high-voltage power supply (HVPS) and with a motor attached to run through the wavelength region set to scan. Aligned in front of the entrance slit of the spectrometer there was positioned the source light needed for the each part of the experiment that was being performed at the given moment. On the other side, attached to the exit slit, there was a photomultiplier tube (PMT) attached to the exit slit converting the light into readable voltage signal which was detected by the NI USB-6008 and transferred to the computer for the analysis of the data.

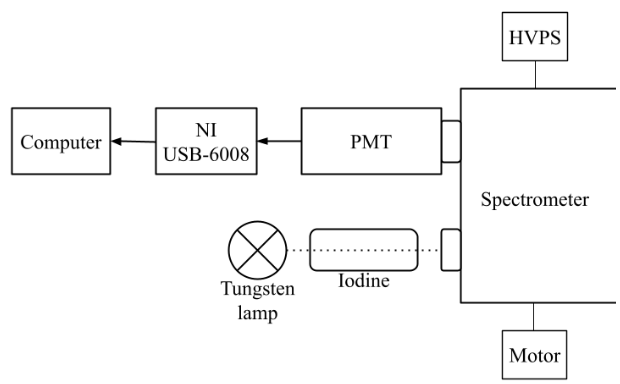

Figure 5: Schematic of the apparatus set-up. With a *Baird & Watts Monospek 1000* spectrometer, a motor, a high-voltage power supply (HVPS), a tungsten lamp and an iodine cell, a photomultiplier tube (PMT) and a National Instruments USB-6008/6009 connected to the computer.

The sources used for calibration were: a red He-Ne laser, a mercury lamp and a sodium lamp. Once the laser was aligned with the entrance slit, leaving enough space between the two, the laser was used to position and align the rest of the sources.

3.2 Experimental procedure

3.2.1 Calibration

The widths of the entrance and exit slits along with the speed of the motor were adjusted until a good enough signal in the 400-650 nm range could be obtained. After this, the calibration was started; recording the spectra of the calibration sources from 650 through 400 nm and cross-matching the observed peak positions of the emission lines with their known wavelengths[5].

For each source, the voltage of the HVPS was adjusted accordingly so that a strong signal could be obtained without surpassing the maximum 10 V reading signal set by the NI USB-6008. In addition, since the experiment relies on turning on the motor and start the recording of the readings through the computer at the exact same time, the spectrum of each source was measured at least 5 times. Then an average of the bandhead positions was taken with the goal to improve the reproducibility of the experiment.

Once the positions of the bandheads of the spectra were acquired and linked to their wavelengths, a linear relationship $\lambda = (x - c)/m$ between the two was obtained. Where x is the position and the parameters m and c are the slope and intercept of the straight line. These were later used to convert into wavelengths the x-axis of the spectra.

3.2.2 Iodine molecular spectrum and the Birge-Sponer plot

The spectrum of the tungsten lamp with and without iodine was recorded multiple times since, again, the experiment relies on starting the motor and the recording of the readings at the same time. From these recordings, a pair of *with* and *without* iodine spectra with very similar delay in the start of the recordings (and therefore very good overlapping) was selected for the experiment.

A wavelength offset was applied to cancel this delay. This was done by examining the obtained spectrum with iodine and comparing it to the spectrum shown in Figure 3, cross-matching the absorption lines to calculate the offset that was necessary to apply.

Once this was calculated and applied, $\log(I_0/I)$ was plotted against the wavelengths λ. Where I and I_0 represent respectively the signal for the spectrum of the tungsten line with and without iodine.

The peaks due to the vibrational series $v'' = 0$ were numbered according to Figure 3 and used to calculate the first differences $\Delta^1 G(v')$ in cm^{-1} which were plotted against the vibrational quantum numbers v' to create a Birge-Sponer plot.

From there, parameters like ω'_e, $\omega'_e X'_e$, v'_m, D'_e, D'_0 and k'_e were calculated as detailed in section 2.3. Numerical calculations of those constants were also performed for comparison following the equations in section 2.2. These were compared to literature values from Herzberg[4] calculating the difference between them.

Finally, these parameters were used to calculate and plot a representation of the Morse potential as a function for internuclear distance r.

This process was attempted to be repeated for the transitions ($v' \leftarrow 1''$). However, the signal was not clear enough to detect and identify enough spectral bands belonging to this vibrational series to continue with the analysis (see Appendix).

4 Results and discussion

The calibration line shown in Figure 6 was obtained, showing the linear relationship between iterations and wavelength with a slope of -12.66 ± 0.11 and an intercept at 8231.70 ± 68.66.

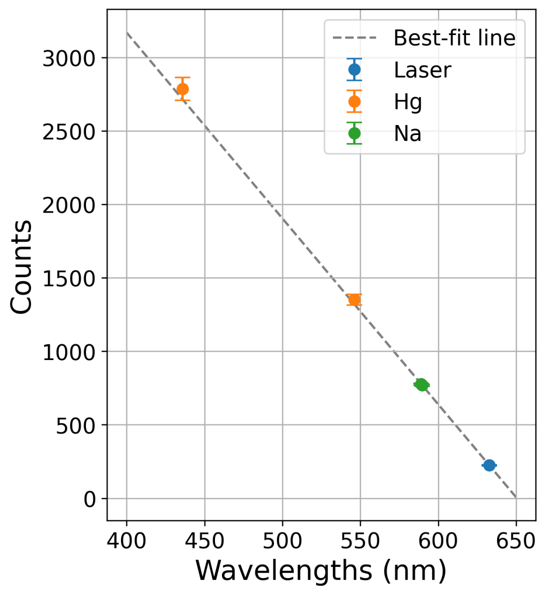

Figure 6: Calibration line for iterations to wavelengths conversion using an He-Ne red laser, a mercury and a sodium lamp.

It can be observed how the shorter the wavelengths, the bigger the errors. This is because the scan was made starting at 650 nm and decreasing in wavelength and the longer the spectrometer had been running for, the larger the possibility of the spectrometer skipping some wavelengths will be. This was consistent throughout the rest of the experiment as all spectrums were measured starting at 650.0 nm.

Although more (but weaker) spectral lines could have been added into the calculation of the calibration line, it was considered that the line was good enough with 5 data points as it goes close to (if not through) the error window of all points involved and χ^2's p-value for the linear fit was 0.644 indicating the fit is valid[6].

4.1 Visualization of the iodine spectrum

A wavelength offset of 9.91 nm was calculated and applied as described in section 3.2.2 with a statistical uncertainty of 0.19 nm (see Appendix for the process of comparison and determination of the offset). The semi-log plot obtained after the offset was applied can be seen in Figure 7 showing the regions described by Figure 2 including the convergence limit, where $(v', 1'')$ series is noticeable and where the near-infrared bands would start appearing.

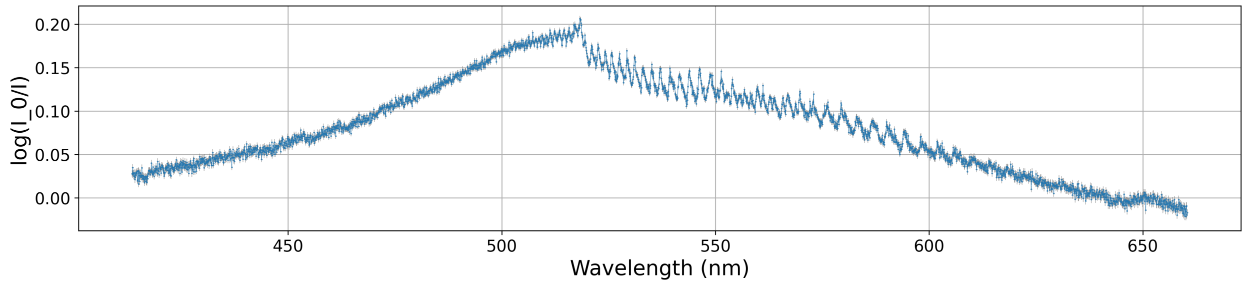

Figure 7: Semi-log plot obtained from plotting the log of the spectrum of the tungsten lamp against the the spectrum of the transmitted light of the tungsten lamp through the cell of iodine vapour.

4.2 The Birge-Sponer plot

With the absorption lines numbered and the first differences calculated, the Birge-Sponer plot seen below was be obtained.

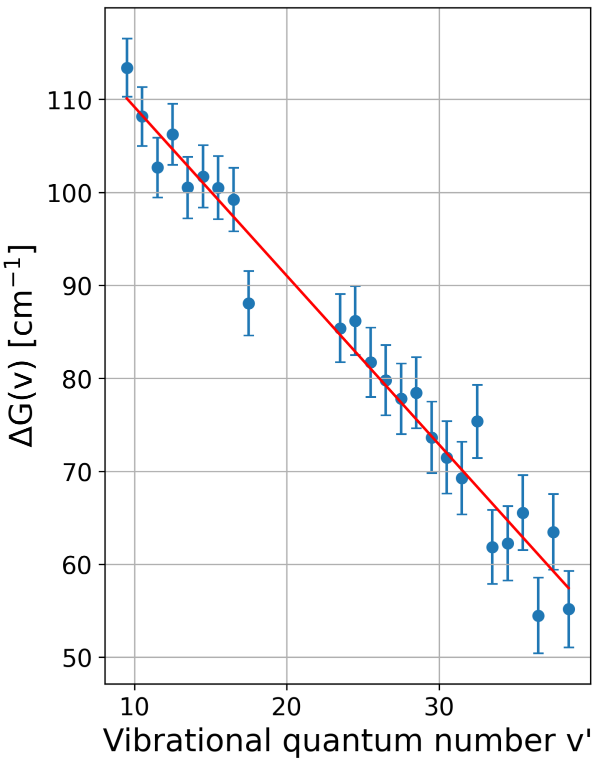

Figure 8: Birge-Sponer plot showcasing the linear relationship between the differences $\Delta^1 G(v')$ and the vibrational quantum numbers v' for $v' < 50$.

In Figure 8 vibrational quantum numbers between 9 and 40 show a linear relationship between the first difference and the vibrational quantum numbers where $\Delta^1 G(v')$ decreases as v' increases. Bands past $v' = 40$ could not be accurately included without sacrificing the goodness of the fit as the bands started being less sharp and the errors started being greater (as they correspond to shorter wavelengths). Absorption lines representing $18 > v' > 23$ (corresponding to section 2 in Figure 2) were numbered but not plotted as it was difficult to distinguish them to the transitions corresponding to other vibrational series.

With the vibrational quantum numbers $v' > 40$ not being included, it was not possible to observe the deviation from linearity achieved around $v' > 50$[2]. Despite this, a good linear fit[6] was obtained (χ^2 p-value of 0.683) and an accurate calculation of the molecular constants were achieved (see Table 1).

The deviation of the obtained values from the ones provided in literature[4] can be seen tabulated. The large difference on $\omega_e' X_e'$ might be due to its small value and therefore high sensitivity to noise, errors and peak misalignment.

	Graphical ($\mathbf{cm^{-1}}$)	**Numerical** ($\mathbf{cm^{-1}}$)	**Accepted** ($\mathbf{cm^{-1}}$)[3][4]	**Difference** (%)
ω_e'	128.3 ± 1.9	142.5 ± 1.4	128.0	0.23
$\omega_e' X_e'$	0.908 ± 0.039	1.214 ± 0.109	0.834	8.87
D_e'	4497 ± 213	4179 ± 377	4398	2.25
D_0'	4432 ± 213	4108 ± 377	4335	2.24

Table 1: Table containing results obtained through the Birge-Sponer plot, results obtained numerically and literature values from Herzberg[4] and Fred E. Stafford[3]. Last column indicates the difference in percentage between the graphical results and the accepted values.

It is also important to note that the values obtained through the Birge-Sponer plot were closer to the accepted literature values than those obtained through numerical methods and often with larger uncertainties. Again, this might be due to the propagation of errors and the higher sensitivity of these through numerical methods (specially the sensitivity to errors and deviation on the seconds differences $\Delta^2 G(v')$ - used in through all of the numerical calculations but none of the graphical ones) compared to the *smoothing* of the noise through the best fit line.

The force constant was calculated to be $k'_e = 6.146 \times 10^4 \pm 0.18 \times 10^4$ dyn/cm, which, although it does not agree, it is in a very close order of magnitude as the literature value of $k'_e = 1.72 \times 10^5$ by Herzberg[1][4]. The vibrational quantum number v_m was found to be $v'_m \approx 70 \pm 3$ which is lower than the last bound value found by LeRoy and Bernstein of $v' = 87$[2]. The difference in these values is due to the fact that the value $v'_m = 70$ was calculated by extrapolating the straight fit line and therefore didn't take into account the non-linearity for values $v' > 50$.

4.3 The Morse potential

With these constants having been calculated, the Morse potential for the electronic state B was calculated and plotted as it can be seen Figure 9. This can be compared to Figure 1 to analyse the true energies at the different vibrational quantum numbers $v' < 50$ (including $G(0')$), where the linearity is maintained. The decrease in the first differences as v' increases until dissociation is reached at D'_e can also be noted as well as the minimum in potential energy $U(r - r_e)$ at the equilibrium internuclear distance r_e.

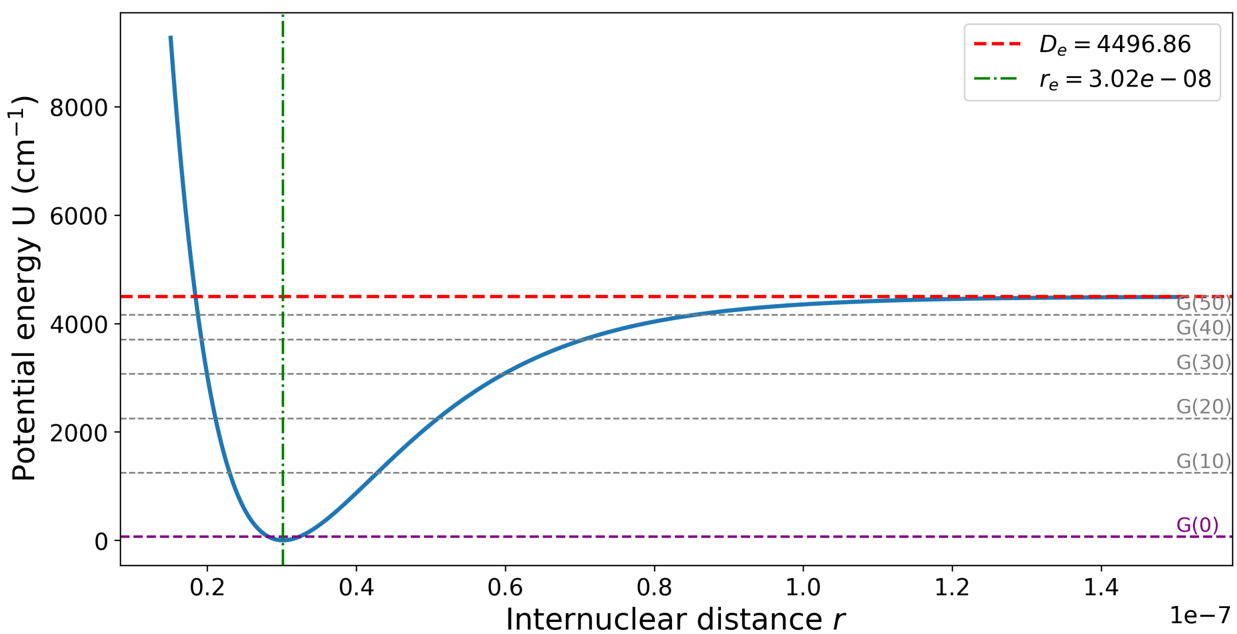

Figure 9: Morse potential plot of I_2's electronic state B showcasing the dissociation energy as well as the vibrational energy for quantum numbers $v' < 50$ where linearity is conserved.

5 Conclusion

With the data obtained from this experiment, the band structure of the diatomic molecule iodine was successfully studied, observing a linear relationship between the first differences of the bands and the vibrational quantum numbers for $v < 50$ in the Birge-Sponer plot. A plot that showed a fundamental vibrational frequency of $128.3 \pm 1.9\,\mathrm{cm}^{-1}$ and an anharmonicity constant of $0.908 \pm 0.039\,\mathrm{cm}^{-1}$ as well as a dissociation energy D'_e of $4497 \pm 213\,\mathrm{cm}^{-1}$ which can also be appreciated in the plot of the Morse potential obtained for the B electronic state. A force constant of $6.146 \times 10^4 \pm 0.18 \times 10^4$ dyn/cm was calculated as well as a vibrational quantum number $v' = v'_m \approx 70$ which was obtained by extrapolating the straight fit line from the Birge-Sponer plot, causing it to deviate from the last bound value $v' = 87$ found by LeRoy and Bernstein since the bandheads appearing after linearity breaks were not able to be identified. Other than that, all values fall within reasonable agreement with the literature data, confirming the reliability of the method despite its reproducibility.

Although uncertainties were included with each measurement, reproducibility was improved and goodness-of-fit tests were applied where applicable, several sources of uncertainty may have not been fully accounted for. Potential systematic errors include calibration inaccuracies of the spectrometer, wavelength drifts, and small imperfections in the iodine cell such as temperature or pressure variations. Statistical uncertainties might arise from sources such as detector sensitivity fluctuations and background light variations. In addition to this, even though several scans of each source were performed in order to improve reproducibility, small inconsistencies may still affect the overall errors and results.

Further research could include the heating of the iodine cell in order to obtain a better signal of spectral bands due to the $(v' \leftarrow 1'')$ transitions which would allow for the calculations of molecular constants such as ω''_e, $\omega''_e X''_e$ and D''_e ultimately allowing for a proper study of the X and B band of iodine and their Morse potential as a function of internuclear radius. Additionally, performing the experiment using fluorescence spectroscopy instead of - or alongside - absorption would provide more precise measurements of the vibrational bands and reduce the effects of overlapping lines, improving the accuracy of the extracted molecular constants.

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